



Network Topologies

LAN topologies

WAN topologies



LAN topologies

- Physical
 - Describes the geometric arrangement of components that make up the LAN
- Logical
 - Describes the possible connections between pairs of networked end-points that can communicate



LAN Topologies(Physical)

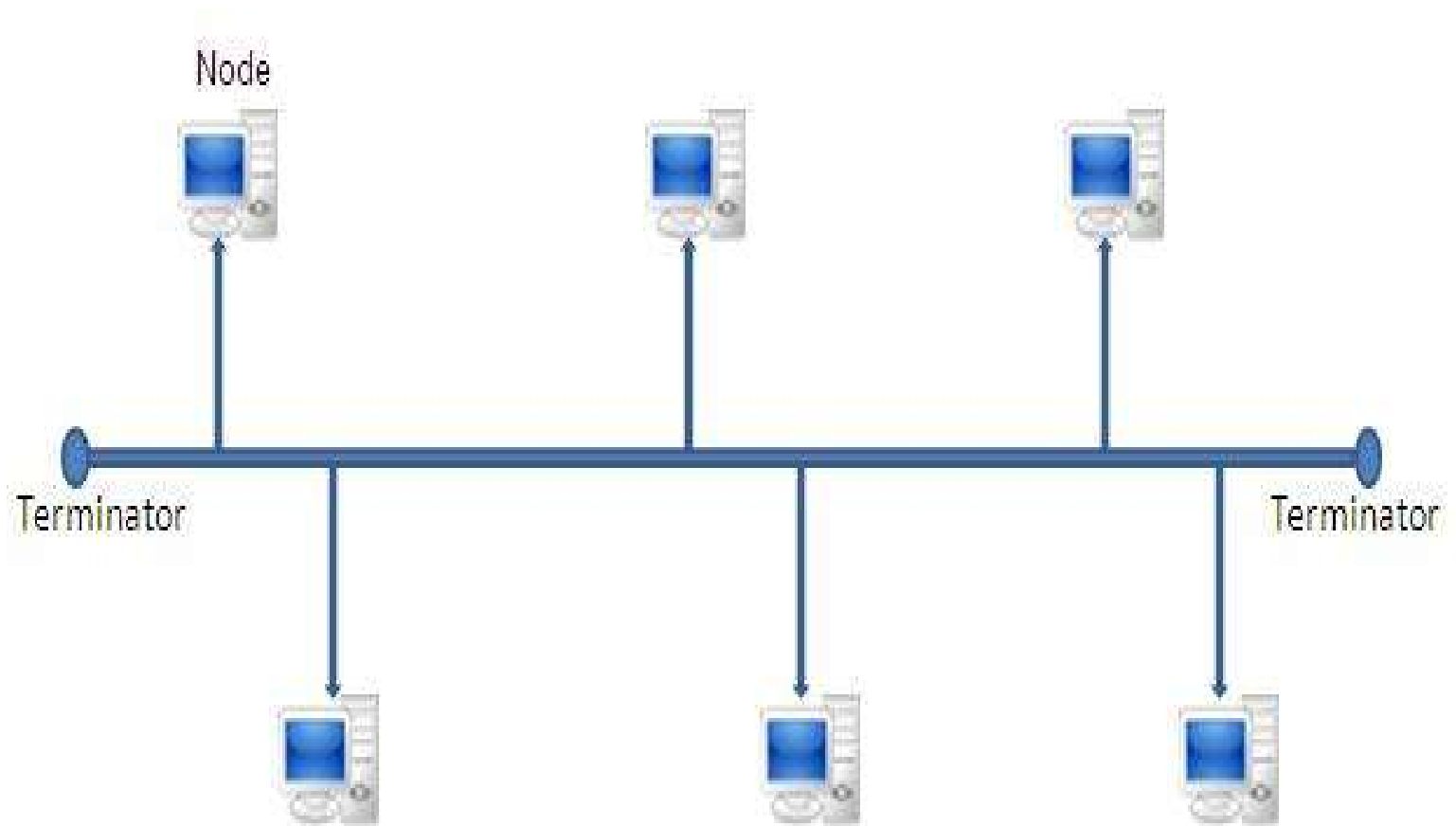
- 1) Bus
- 2) Star
- 3) Ring
- 4) Mesh



Bus topology

- All networked nodes are interconnected, peer to peer, using a single, open-ended cable
- Both ends of the bus must be terminated with a terminating resistor to prevent signal bounce

Bus topology





Advantages of Bus topology

- 1) Easy to implement and extend
- 2) Well suited for temporary networks that must be set up in a hurry
- 3) Typically the least cheapest topology to implement
- 4) Failure of one station does not affect others



Disadvantages of Bus topology

- 1) Difficult to administer/troubleshoot
- 2) Limited cable length and number of stations
- 3) A cable break can disable the entire network; no redundancy
- 4) Maintenance costs may be higher in the long run
- 5) Performance degrades as additional computers are added



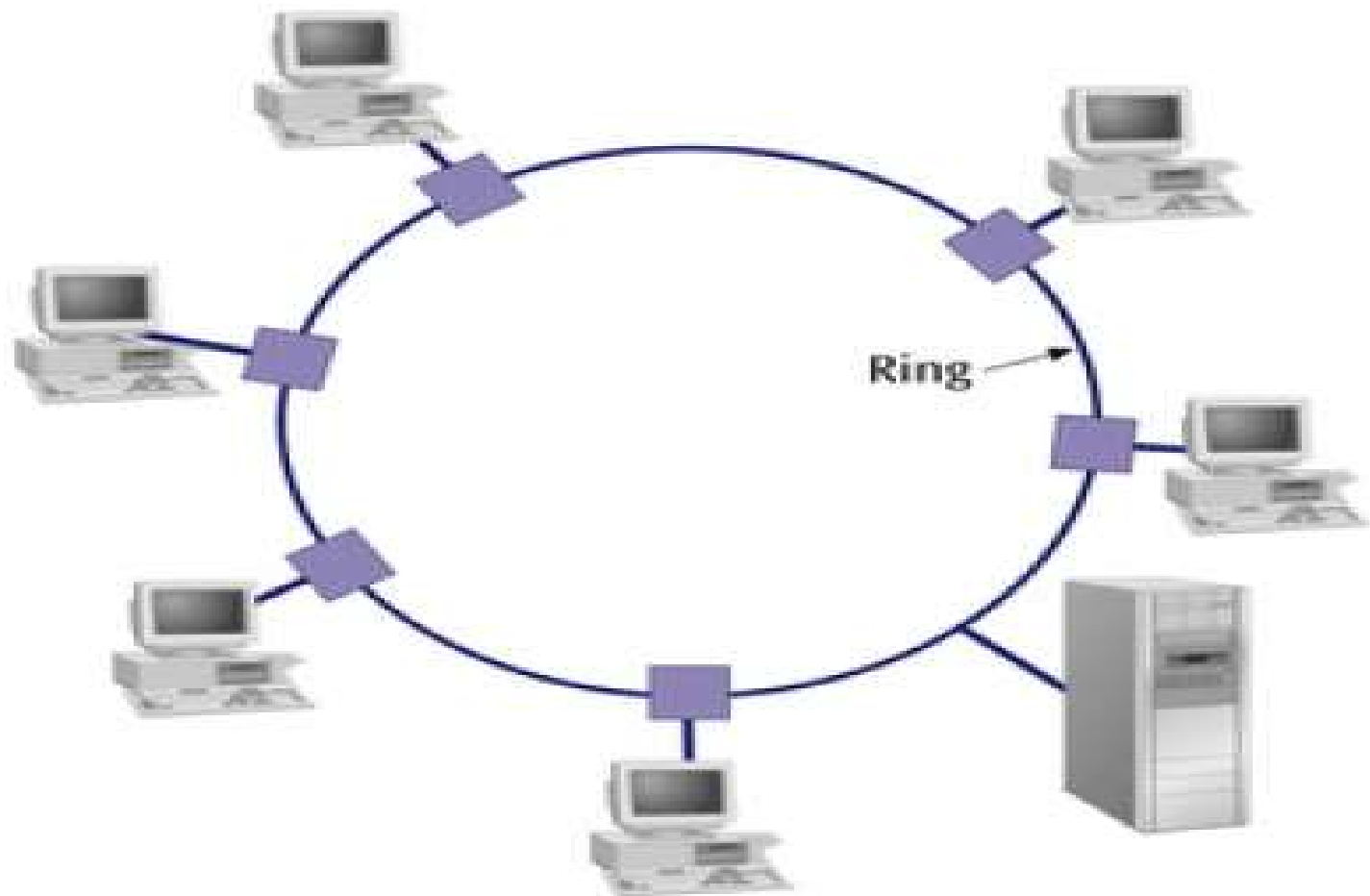
Ring topology

- started out as a simple peer-to-peer LAN topology
- Each networked workstation had two connections: one to each of its nearest neighbors
- Data was transmitted unidirectionally around the ring
- Sending and receiving of data takes place by the help of TOKEN

Token Passing

- Token contains a piece of information which along with data is sent by the source computer
- This token then passes to next node, which checks if the signal is intended to it
 - If yes, it receives it and passes the empty token into the network
 - otherwise passes token along with the data to next node

Ring topology





Advantages of Ring topology

- 1) This type of network topology is very organized
- 2) Performance is better than that of Bus topology
- 3) No need for network server to control the connectivity between workstations
- 4) Additional components do not affect the performance of network
- 5) Each computer has equal access to resources



Disadvantages of Ring topology

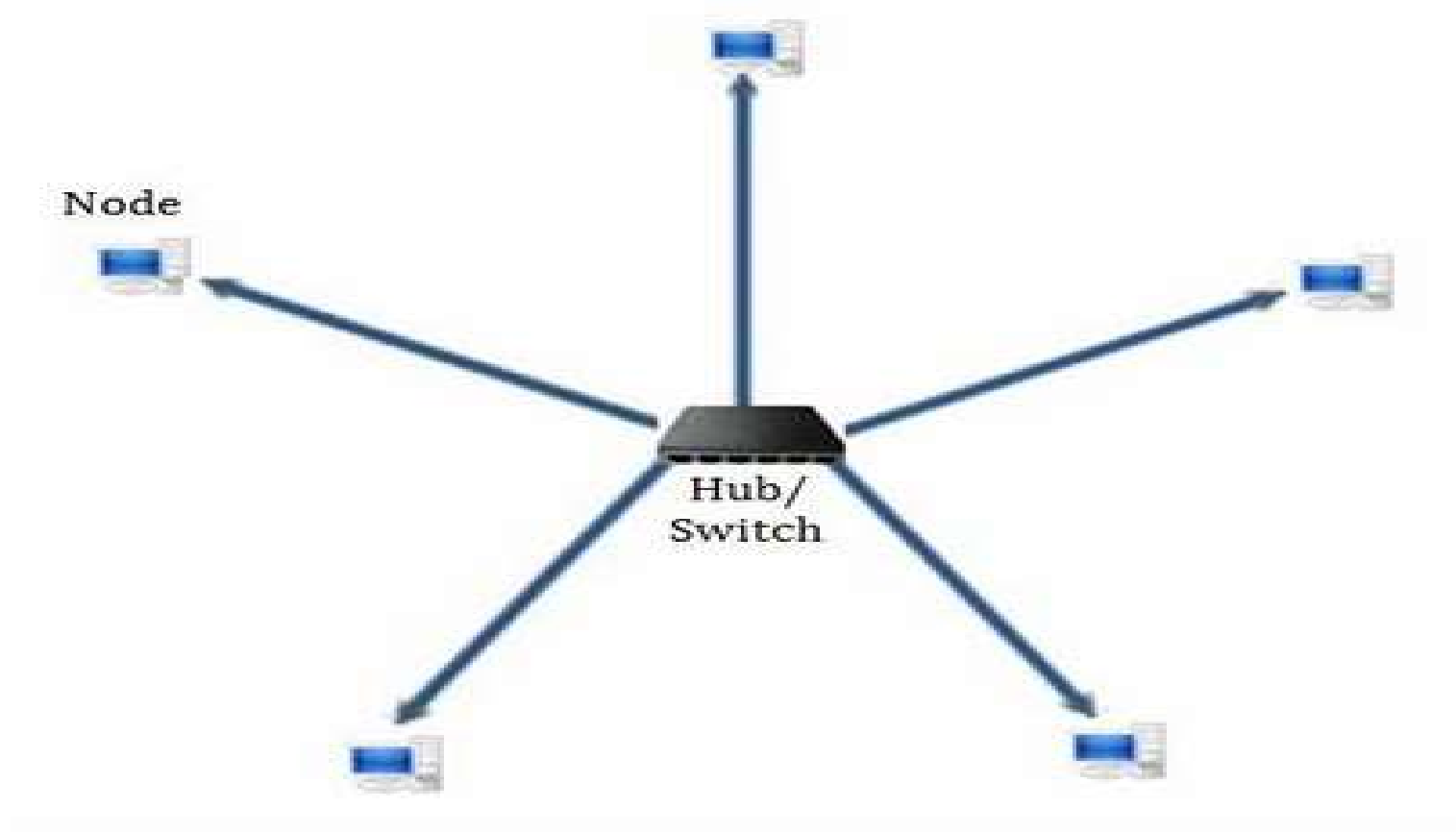
- 1) Each packet of data must pass through all the computers between source and destination, slower than star topology
- 2) If one workstation or port goes down, the entire network gets affected
- 3) Network is highly dependent on the wire which connects different components



Star topology

- Have connections to networked devices that “radiate” out from a common point
- Each networked device in star topology can access the media independently
- Have become the dominant topology type in contemporary LANs
- Stars have made buses and rings obsolete in LAN topologies

Star topology





Advantages of star topology

- 1) Compared to Bus topology it gives far much better performance
- 2) Easy to connect new nodes or devices
- 3) Centralized management. It helps in monitoring the network
- 4) Failure of one node or link doesn't affect the rest of network

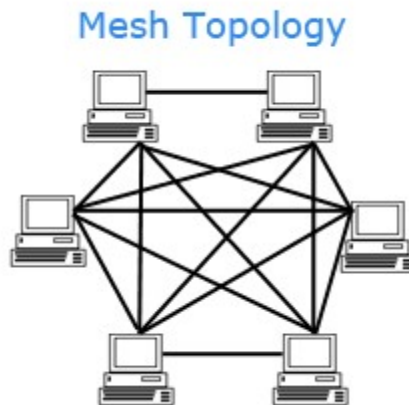


Disadvantages of star topology

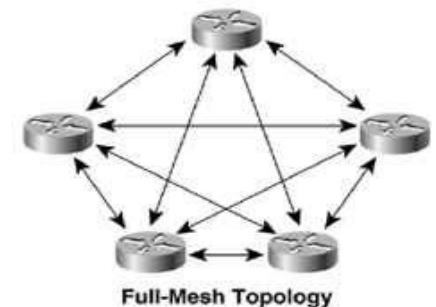
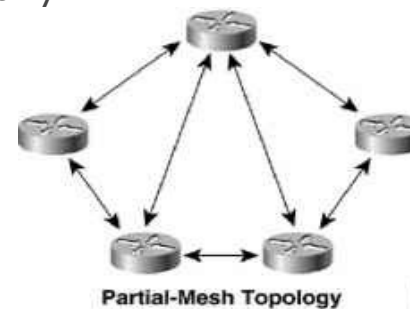
- 1) If central device fails whole network goes down
- 2) The use of hub, a router or a switch as central device increases the overall cost of the network
- 3) Performance and as well number of nodes which can be added in such topology is depended on capacity of central device

Mesh topology

- A mesh topology is a network setup where each computer and network device is interconnected with one another. This topology setup allows for most transmissions to be distributed even if one of the connections goes down. It is a topology
- commonly used for wireless networks.



- There are two forms of this topology: full mesh and a partially-connected mesh.
- In a *full mesh topology*, every computer in the network has a connection to each of the other computers in that network. The number of connections in this network can be calculated using the following formula (n is the number of computers in the network): $n(n-1)/2$
- In a *partially-connected mesh topology*, at least two of the computers in the network have connections to multiple other computers in that network. It is an inexpensive way to implement redundancy in a network. If one of the primary computers or connections in the network fails, the rest of the network continues to operate normally.



- **Advantages of a mesh topology**

- Manages high amounts of traffic, because multiple devices can transmit data simultaneously.
- A failure of one device does not cause a break in the network or transmission of data.
- Adding additional devices does not disrupt data transmission between other devices.

- **Disadvantages of a mesh topology**

- The cost to implement is higher than other network topologies, making it a less desirable option.
- Building and maintaining the topology is difficult and time consuming.
- The chance of redundant connections is high, which adds to the high costs and potential for reduced efficiency.